

## Problem 1

1. Consider:  $\mathbb{P}(X = k)$  the total number of ways to pick  $n$  balls from  $2n$  balls is  $\binom{2n}{n}$ . We then want to count the number of ways to pick  $k$  black balls which is  $\binom{n}{k}$ , and thus we must also pick  $n - k$  blue balls this is counted in  $\binom{n}{n-k}$  thus  $\mathbb{P}(X = k) = \frac{\binom{n}{k}\binom{n}{n-k}}{\binom{2n}{n}}$

2. Let  $X_i$  be 1 if we get a black ball on the  $i$ th spot and zero otherwise.

$$\text{Thus } E[X^2] = E[X_1^2 + X_2^2 + \dots + X_n^2 + 2X_1X_2 + 2X_1X_3 + \dots + 2X_{n-1}X_n] = \sum_{i=1}^n E[X_i^2] + 2 \sum_{1 \leq i < j \leq n} E[X_iX_j]$$

since  $X_i$  only outputs 0 or 1  $X_i^2$  will also output the same thing thus

$$\begin{aligned} &= \sum_{i=1}^n E[X_i] + 2 \sum_{1 \leq i < j \leq n} E[X_iX_j] \\ &= \sum_{i=1}^n P(I_i = 1) + 2 \sum_{1 \leq i < j \leq n} P(I_i = 1, I_j = 1) \\ &= \sum_{i=1}^n \frac{\binom{2n-1}{n-1}}{\binom{2n}{n}} + 2 \sum_{1 \leq i < j \leq n} \frac{\binom{2n-2}{n-2}}{\binom{2n}{n}} \\ &= \sum_{i=1}^n \frac{1}{2} + 2 \binom{n}{2} \frac{n-1}{2(2n-1)} \\ &= \frac{n}{2} + \frac{n(n-1)^2}{2(2n-1)} \\ &= \frac{n((n-1)^2 + 2n-1)}{2(2n-1)} \end{aligned}$$

## Problem 2

Consider  $\mathbb{P}(M \geq k)$  since  $\{M \geq k\} = (\{M \geq K\} \cap \{X < k\}) \cup (\{M \geq K\} \cap \{X = k\}) \cup (\{M \geq K\} \cap \{X > k\})$ , we know this because we can put where the path ends into those three groups above. Then because these sets are disjoint since you can't end at  $k$ , end at greater than  $k$  and end at less than  $k$  all at the same time, so we know.

$$\mathbb{P}(M \geq k) = \mathbb{P}(M \geq k, X = k) + \mathbb{P}(M \geq k, X > k) + \mathbb{P}(M \geq k, X < k)$$

Look at  $\mathbb{P}(M \geq k, X = k)$  since we know that the end is  $k$  we know that the max is at least  $k$ , so thus  $\mathbb{P}(M \geq k, X = k) = \mathbb{P}(X = k)$

Look at  $\mathbb{P}(M \geq k, X > k)$  since we know that the end is greater than  $k$  this is equivalent to  $X \geq k+1$ , and thus we know that the max is at least  $k+1$  already this  $\mathbb{P}(M \geq k, X > k) = \mathbb{P}(M \geq k, X \geq k+1) = \mathbb{P}(X \geq k+1)$

Look at  $\mathbb{P}(M \geq k, X < k)$  since we know the max is at  $M$  we then know that the path goes down to less than  $k$ . Thus, by reflection we can flip up at  $M$  which is a bijection to  $X > k \implies X \geq k+1$  and thus  $\mathbb{P}(M \geq k, X < k) = \mathbb{P}(M \geq k, X > k) = \mathbb{P}(X \geq k+1)$

Thus  $\mathbb{P}(M \geq k) = \mathbb{P}(X = k) + 2\mathbb{P}(X \geq k+1)$

### Problem 3

1. The probability of not getting one in one booster is  $(1 - \frac{15}{300})$  the probability of getting at least one is  $1 - (1 - \frac{15}{300})^n$
2. First order the cards 1-300 then let  $X_i$  be 1 if they get card  $i$  and 0 if not.  $E[X] = E[X_1 + X_2 + \dots + X_{300}] = \sum_{i=0}^{300} E[X_i] = \sum_{i=0}^{300} \mathbb{P}(\text{Getting card } i) = \sum_{i=0}^{300} (1 - (1 - \frac{15}{300})^n) = 300(1 - (1 - \frac{15}{300})^n)$
3. The probability of not getting one in one card is  $(1 - \frac{1}{300})$  the probability of getting at least one is in  $15n$  cards  $1 - (1 - \frac{1}{300})^{15n}$

First order the cards 1-300 then let  $X_i$  be 1 if they get card  $i$  and 0 if not.

$$E[X] = E[X_1 + X_2 + \dots + X_{300}] = \sum_{i=0}^{300} E[X_i] = \sum_{i=0}^{300} \mathbb{P}(\text{Getting card } i) = \sum_{i=0}^{300} (1 - (1 - \frac{1}{300})^{15n}) = 300(1 - (1 - \frac{1}{300})^{15n})$$

$$(\frac{299}{300})^{15} \approx (0.9967)^{15} \text{ and } \frac{285}{300} = 0.95 \text{ thus } (0.9967)^{15} \text{ and } (0.95)^{15} \text{ since the left one is bigger}$$

thus it will go to zero slower than the other one so  $\frac{299^{15n}}{300^{15n}} > \frac{285^{15n}}{300^{15n}}$  thus  $1 - (\frac{299}{300})^{15n} < 1 - (\frac{285}{300})^{15n}$   
 thus  $E[X_{\text{individual}}] < E[X_{\text{boosters}}]$